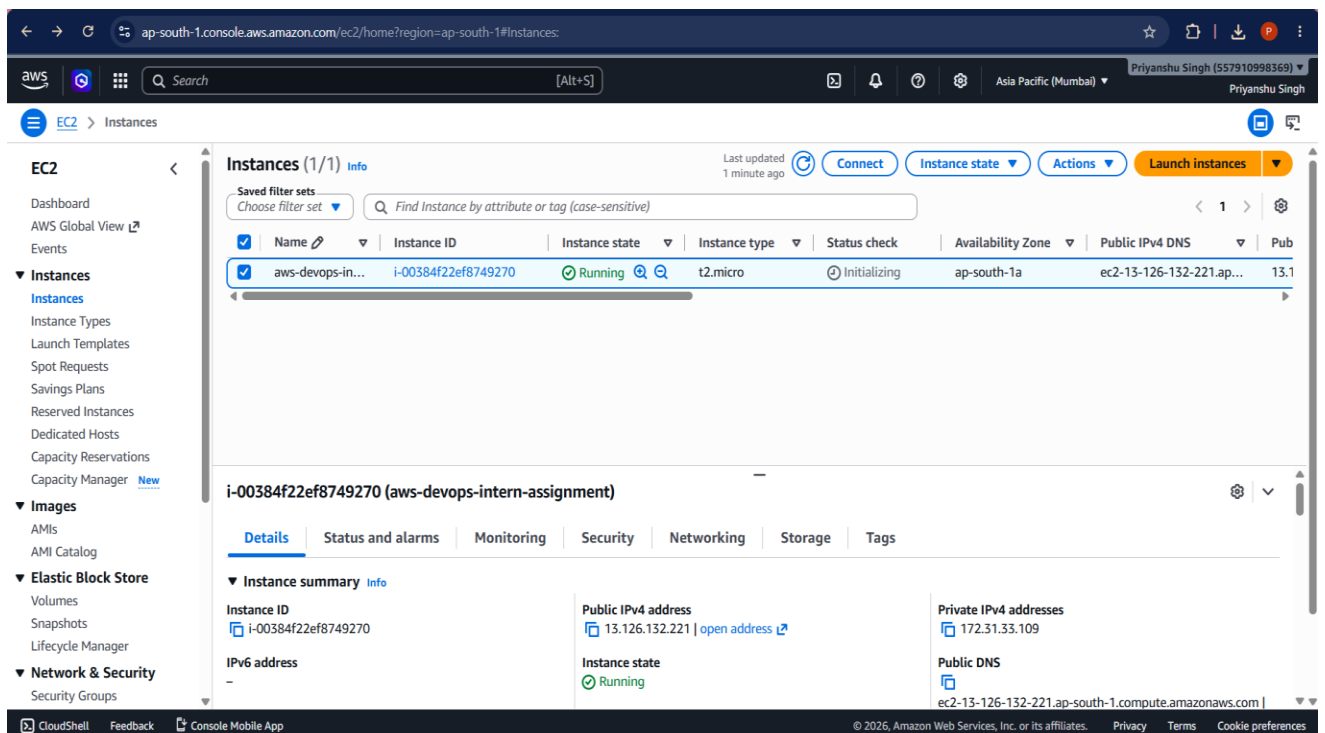


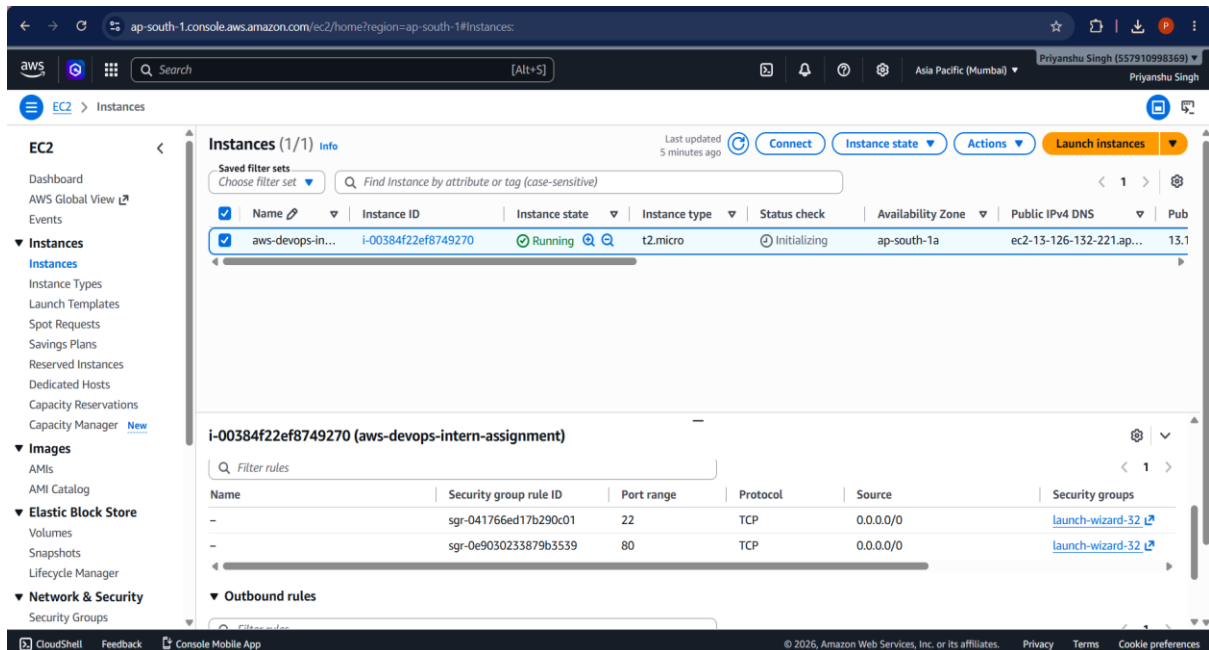
AWS DevOps Engineer Intern Assignment

Objective: Deploy a simple website on an AWS EC2 instance and document the process.

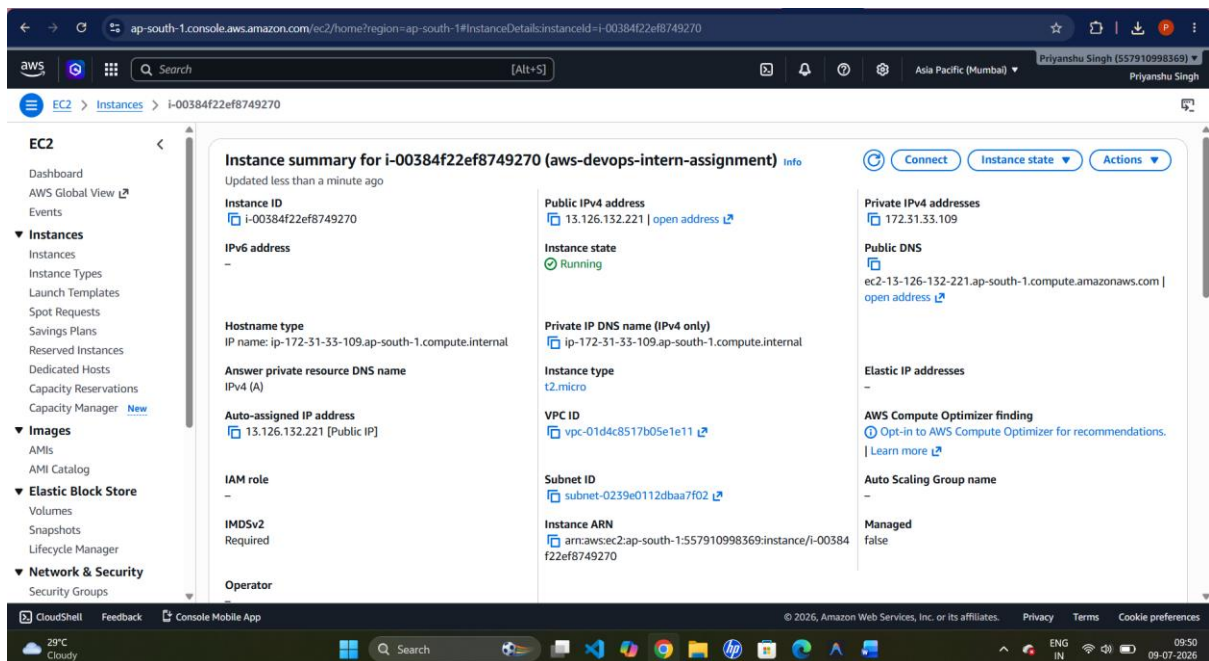
Logged in to the AWS Management Console and launched an Ubuntu 22.04 LTS EC2 instance. A Security Group was configured to allow SSH (Port 22) and HTTP (Port 80) traffic. The instance was successfully launched and assigned a public IP address.



Configured inbound rules in the Security Group to allow SSH access for remote administration and HTTP access so the hosted website can be accessed through a web browser.



Details of the Instance



Connected securely to the Ubuntu EC2 instance using SSH and the generated AWS key pair.

```
ubuntu@ip-172-31-33-109: ~  
C:\Users\hpn>cd Downloads  
C:\Users\hpn\Downloads>ssh -i "intern.pem" ubuntu@ec2-13-126-132-221.ap-south-1.compute.amazonaws.com  
The authenticity of host 'ec2-13-126-132-221.ap-south-1.compute.amazonaws.com (13.126.132.221)' can't be established.  
ED25519 key fingerprint is SHA256:6hKk5Ks9aztV73uep9iEpl/ao+G3DI1DUWaxp8zN3IY.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'ec2-13-126-132-221.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Thu Jul  9 04:21:29 UTC 2026  
  
System load:  0.3          Processes:      113  
Usage of /:   30.6% of 6.61GB  Users logged in: 0  
Memory usage: 24%          IPv4 address for enx0: 172.31.33.109  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
ubuntu@ip-172-31-33-109:~$ |
```

Installed the Nginx web server, which will be used to host the static HTML website.

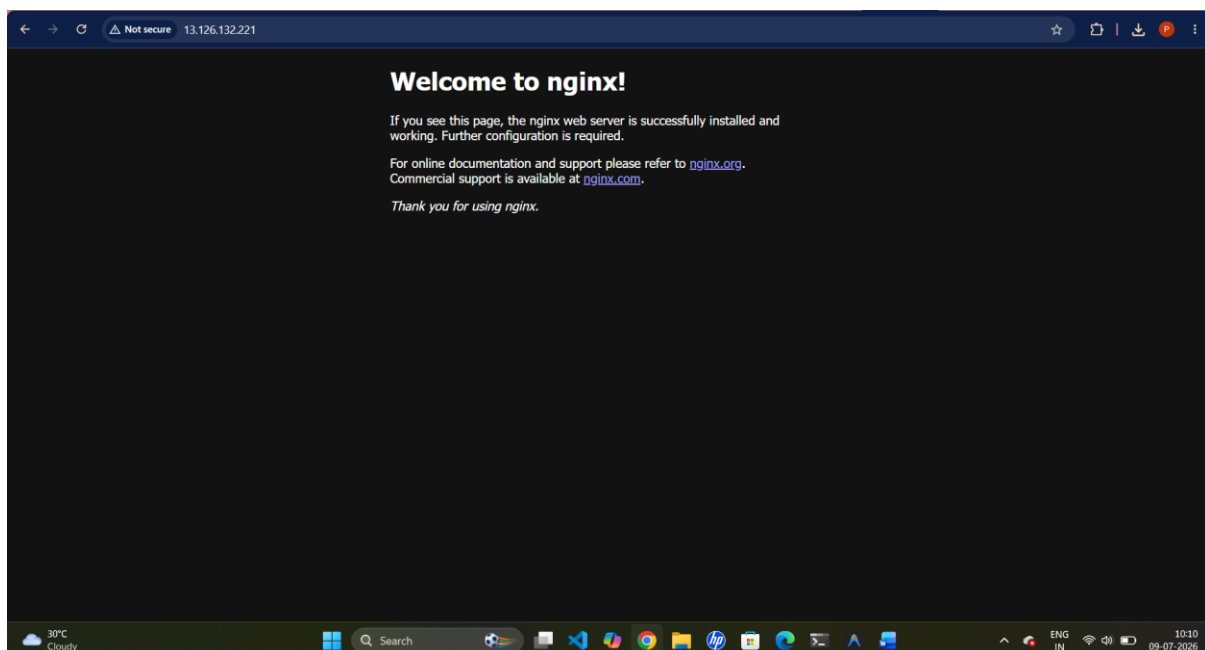
Verified that the Nginx service was running successfully and restarted the service to ensure all configurations were applied correctly.

```
ubuntu@ip-172-31-33-109:~$ sudo systemctl status nginx  
● nginx.service - A high performance web server and a reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)  
   Active: active (running) since Thu 2026-07-09 04:27:13 UTC; 1min 29s ago  
     Invocation: fb964b637d5647ad8eb185b614b7143b  
       Docs: man:nginx(8)  
    Process: 1667 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
    Process: 1669 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
   Main PID: 1697 (nginx)  
      Tasks: 2 (limit: 657)  
     Memory: 2.3M (peak: 5.3M)  
        CPU: 32ms  
   CGroup: /system.slice/nginx.service  
           └─1697 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"  
             └─1700 "nginx: worker process"  
  
Jul 09 04:27:13 ip-172-31-33-109 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...  
Jul 09 04:27:13 ip-172-31-33-109 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.  
ubuntu@ip-172-31-33-109:~$
```

Checked the available disk space and memory usage of the EC2 instance to monitor basic system resources.

```
ubuntu@ip-172-31-33-109: ~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-172-31-33-109:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.3G  4.4G  35% /
tmpfs            476M   0  476M   0% /dev/shm
tmpfs            191M  980K  190M   1% /run
tmpfs            476M   0  476M   0% /tmp
none            1.0M   0   1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0   1.0M   0% /run/credentials/systemd-resolved.service
/dev/xvda13      980M  96M  827M  11% /boot
/dev/xvda15      105M  6.3M   99M   7% /boot/efi
none            1.0M   0   1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M   0   1.0M   0% /run/credentials/getty@tty1.service
none            1.0M   0   1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs            96M   0   96M   0% /run/user/1000
ubuntu@ip-172-31-33-109:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           951Mi       327Mi       190Mi       1.8Mi       552Mi       624Mi
Swap:            0B           0B           0B
```

Accessed the EC2 Public IP through a web browser to verify that the Nginx web server was successfully serving the default webpage.



Replaced the default Nginx landing page with a custom HTML webpage containing personal details as required in the assignment.

```
ubuntu@ip-172-31-33-109: /var/www/html$ cat index.html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>AWS DevOps Internship Assignment | Priyanshu Singh</title>

<style>

*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:Segoe UI,Tahoma,Geneva,Verdana,sans-serif;
}

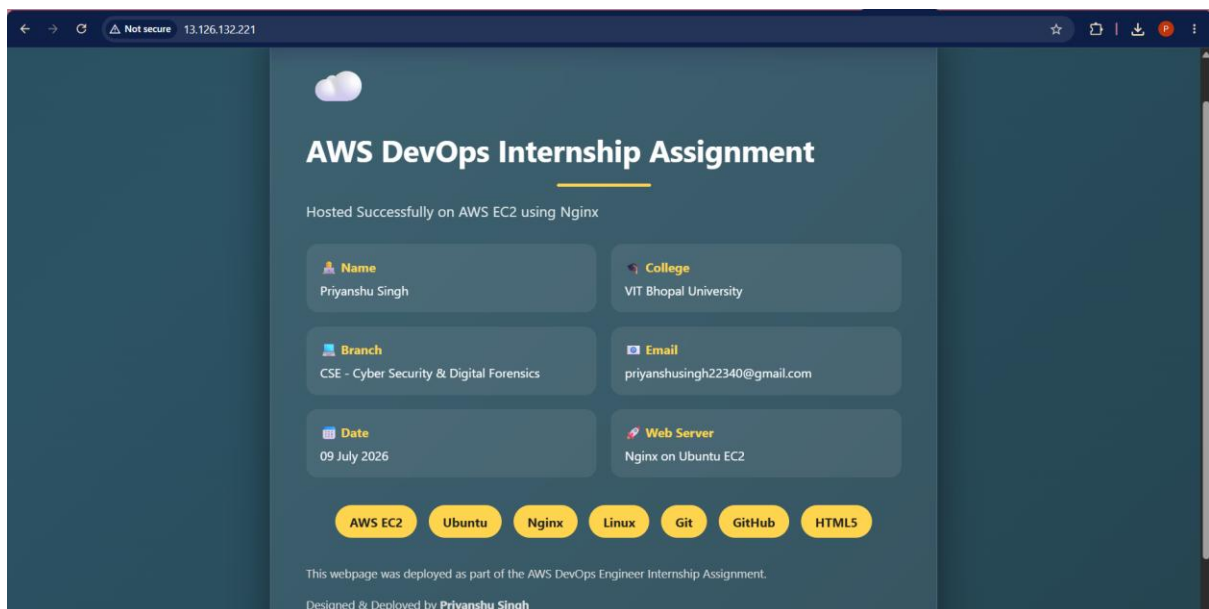
body{
background:linear-gradient(-45deg,#0F2027,#203A43,#2C5364,#1d4350);
background-size:400% 400%;
animation:bg 12s ease infinite;
min-height:100vh;

display:flex;
justify-content:center;
align-items:center;

padding:30px;
color:white;
}

@keyframes bg{
0%{background-position:0% 50%;}
50%{background-position:100% 50%;}
100%{background-position:0% 50%;}
}
```

Successfully deployed the custom HTML webpage on the EC2 instance and verified that it was accessible through the public IP address.



Installed Docker on the EC2 instance and verified the installation by running the official hello-world container, confirming that Docker was functioning correctly.

```
ubuntu@ip-172-31-33-109: ~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55886f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:96498ffd522e70807ab6384a5c0485a79b9c7c08ca79ba08623edcad1054e62d
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

ubuntu@ip-172-31-33-109:~$ |
```